Introduction to Logic (CS) 2024
Homework Assignment 1

Submission deadline: Tue 17 Sep 2024, 11:00

This homework assignment consists of 6 questions, plus a bonus question. Each question indicates how many points it is worth. With the regular questions, you can earn 100 points. With the bonus question, you can earn additionally 10 points. The grade is the number of points you earned divided by 10, with a maximum of 10.

The submissions instructions are in BrightSpace (Content → Homework Assignments → General Information), **please read them carefully and in full**.

This is an individual assignment. Do not copy your homework from someone (or something) else (this includes LLMs), and do not share your solutions with anyone! If fraud is suspected (e.g., if your answers are too similar compared to the answers of someone else) the Board of Examiners will be notified.

Questions

1. (10 points) For each of the two following arguments, say whether it is valid or not valid. Briefly explain your answer. (Note: You are *not* asked to provide a translation into FOL in this question.)

   (a)
   Audrey likes soup and broccoli.
   Everyone who likes broccoli also likes spaghetti.
   ∴ Someone likes both soup and spaghetti.

   (b)
   No one who likes sloople also likes trimpets.
   Somebody likes bluples and trimpets.
   ∴ Somebody likes sloople and bluples.

   Solution: The first argument is valid. Because Audrey likes broccoli (part of premise 1), and everyone who likes broccoli also likes spaghetti (premise 2), we can conclude that Audrey likes spaghetti (as well as broccoli and soup). It then follows that someone—Audrey—likes both soup and spaghetti.

   The second argument is not valid. A counterexample is given by the following situation. Consider the situation, in which there is only one individual in the world, Jorge. And Jorges like bluples and trimpets, but hates sloople. Since Jorge is the only person in the world, the first premise is satisfied. And since Jorge is someone, the second premise is also satisfied. The conclusion, however, is not satisfied, because there is no person who like sloople and bluples.

2. (6 points) Translate the *propositional logic* formula $P \vee (\neg Q \wedge R)$ into natural sounding English using the following translation key:

   P: It is raining outside.

   Q: I want to do my homework.

   R: I want to play football.

Solution: Several solutions possible. For example:

- It is (either) raining outside, or I do not want to do my homework and I want to play football.
- Unless it is raining outside, I do not want to do my homework, but I want to play football instead.

3. (24 points) Translate the following sentences into *propositional logic*, where atomic sentences are represented by uppercase letters. Also provide the translation key. Make sure that your atomic sentences cannot be broken down further.

   (a) You can call the elevator, or you can take the stairs and get some exercise.

   Solution: $C \vee (S \wedge E)$
   where

   C: You can call the elevator.

   S: You can take the stairs.

   E: You can get some exercise.

   (b) Unless the weather is good, I do not want to picnic with Jesse, but I do want to picnic with Sam regardless of the weather.

   Solution: $(W \vee \neg J) \wedge S$
   where

   W: The weather is good.

   J: I want to picnic with Jesse.

   S: I want to picnic with Sam.

   (c) The movie was great although it was neither funny nor action-packed.

   Solution: $G \wedge \neg(F \vee A)$ or $G \wedge \neg F \wedge \neg A$
   where

   G: The movie was great.

   F: The movie was funny.

   A: The movie was action packed.

4. (6 points) Translate the following *first-order logic* sentence:

$$P(k, v) \vee \big(v = f(k) \wedge \neg C(v)\big)$$

into natural sounding English. The domain of discourse is all people, and the translation key is as follows:

constant k: Keith;

constant v: Valerie;

function $f(x)$: the coworker of x;

unary predicate $C(x)$: x has a fan;

binary predicate $P(x, y)$: x steals a fan from y.

Solution: Several solutions possible. For example:

- (Either) Keith steals a fan from Valerie, or Valerie is the coworker of Keith and Valerie does not have a fan.
- Keith steals a fan from Valerie, or Valerie is the coworker of Keith and (she) does not have a fan.

5. (24 points) Translate the following sentences into *first-order logic*. Also provide the translation key. The domain of discourse is the collection of all people. Make sure that your atomic sentences and your names cannot be broken down further.
   (a) Willem-Alexander is the father of Amalia and Alexia, and he is married to Máxima.
   (b) Amalia and Alexia both study at university, but only one of them studies Politics.
   (c) Alexia and Ariane are both princesses, but neither of them is the Princess of Orange unless Amalia steps down.

Solution:

(a) $(wa = \mathit{father}(am)) \wedge (wa = \mathit{father}(al)) \wedge \mathit{Married}(wa, ma)$.

(b) $\mathit{StudiesAtUni}(al) \wedge \mathit{StudiesAtUni}(am)$
$\wedge \big(\mathit{StudiesPol}(al) \vee \mathit{StudiesPol}(am)\big)$
$\wedge \neg\big(\mathit{StudiesPol}(al) \wedge \mathit{StudiesPol}(am)\big)$.

(c) $\mathit{Princess}(al) \wedge \mathit{Princess}(ar)$
$\wedge \big(\mathit{SD}(am) \vee (\neg \mathit{PO}(al) \wedge \neg \mathit{PO}(ar))\big)$.

with the following translation key:

- constants wa: Willem-Alexander, ma: Máxima.
- constants am: Amalia, al: Alexia, ar: Ariane.
- predicate $\mathit{StudiesAtUni}(x)$: x studies at university.
- predicate $\mathit{StudiesPol}(x)$: x studies politics.
- predicate $\mathit{Princess}(x)$: x is a princess.
- predicate $\mathit{PO}(x)$: x is the Princess of Orange.
- predicate $\mathit{SD}(x)$: x steps down.
- function $\mathit{father}(x)$: the father of x.
- relation $\mathit{Married}(x, y)$: x is married to y.

Solutions where "father of" is translated with a relation will also be considered correct:

- relation $\mathit{Father}(x, y)$: x is the father of y

The translation of (a) is then: $\mathit{Father}(wa, am) \wedge \mathit{Father}(wa, al) \wedge \mathit{Married}(wa, ma)$

6. (30 points) Give formal proofs, including justifications, for the following inferences. You can only use the Reiteration rule and the Introduction and Elimination rules for identity and the Boolean connectives.

(a)
$\neg(\neg A \wedge B)$
B
$\vdash$
A

Solution:

1	$\neg(\neg A \wedge B)$	
2	B	
3	$\quad \neg A$	
4	$\quad \neg A \wedge B$	$\wedge$ Intro: 3, 2
5	$\quad \bot$	$\bot$ Intro: 4, 1
6	$\neg\neg A$	$\neg$ Intro: 3–5
7	A	$\neg$ Elim: 6

(b)
$(a = b) \wedge (c = a)$
$\neg R(a, b)$
$\vdash$
$\neg R(b, c)$

Solution:

1.	$(a = b) \wedge (c = a)$	
2.	$\neg R(a, b)$	
3.	$a = b$	$\wedge$ Elim: 1
4.	$c = a$	$\wedge$ Elim: 1
5.	$c = b$	$=$ Elim: 4,3
6.	$c = c$	$=$ Intro
7.	$b = c$	$=$ Elim: 6,3
8.	$\neg R(b, b)$	$=$ Elim: 2,3
9.	$\neg R(b, c)$	$=$ Elim: 8,7

Another solution:

1.	$(a = b) \wedge (c = a)$	
2.	$\neg R(a, b)$	
3.	$a = b$	$\wedge$ Elim: 1
4.	$c = a$	$\wedge$ Elim: 1
5.	$c = b$	$=$ Elim: 4,3
6.	$a = a$	$=$ Intro
7.	$b = a$	$=$ Elim: 6,3
	8. $R(b, c)$	
	9. $R(a, c)$	$=$ Elim: 8,7
	10. $R(a, b)$	$=$ Elim: 9,5
	11. $\bot$	$\bot$ Intro: 10,2
12.	$\neg R(b, c)$	$\neg$ Intro: 8-11

(c)
$\neg(P(a) \vee Q(b)) \wedge P(b)$
$(a = b) \vee Q(b)$
$\vdash$
$Q(b)$

Solution:

1	$\neg(P(a) \vee Q(b)) \wedge P(b)$	
2	$(a = b) \vee Q(b)$	
3	$P(b)$	$\wedge$ Elim: 1
4	$\quad a = b$	
5	$\quad a = a$	$=$ Intro
6	$\quad b = a$	$=$ Elim: 5, 4
7	$\quad P(a)$	$=$ Elim: 3, 6
8	$\quad \neg(P(a) \vee Q(b))$	$\wedge$ Elim: 1
9	$\quad P(a) \vee Q(b)$	$\vee$ Intro: 7
10	$\quad \bot$	$\bot$ Intro: 9, 8
11	$\quad Q(b)$	$\bot$ Elim: 10
12	$\quad Q(b)$	
13	$\quad Q(b)$	Reit: 12
14	$Q(b)$	$\vee$ Elim: 2, 4–11, 12–13

7. (10 points) (Bonus question) Give a formal proof, including justifications, for the following inference. You can only use the Reiteration rule and the Introduction and Elimination rules for identity and the Boolean connectives.

$\neg P(a, b) \vee \neg P(b, a)$
$(b = a) \vee (b = c)$
$P(c, a)$
$\therefore \neg P(a, b)$

Solution:

1	$\neg P(a, b) \vee \neg P(b, a)$	
2	$(b = a) \vee (b = c)$	
3	$P(c, a)$	
4	$\quad \neg P(a, b)$	
5	$\quad \neg P(a, b)$	Reit: 4
6	$\quad \neg P(b, a)$	
7	$\quad\quad b = a$	
8	$\quad\quad b = b$	$=$ Intro
9	$\quad\quad a = b$	$=$ Elim: 8, 7
10	$\quad\quad \neg P(a, a)$	$=$ Elim: 6, 7
11	$\quad\quad \neg P(a, b)$	$=$ Elim: 10, 9
12	$\quad\quad b = c$	
13	$\quad\quad \neg P(c, a)$	$=$ Elim: 6, 12
14	$\quad\quad \bot$	$\bot$ Intro: 3, 13
15	$\quad\quad \neg P(a, b)$	$\bot$ Elim: 14
16	$\quad \neg P(a, b)$	$\vee$ Elim: 2, 7–11, 12–15
17	$\neg P(a, b)$	$\vee$ Elim: 1, 4–5, 6–16